INDIAN INSTITUTE OF MANAGEMENT ROHTAK

IT Applications Assignment

Hybrid Machine Learning for Hate Speech Detection

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1. Introduction

The rise of social media platforms has transformed global communication, facilitating massive information exchange across diverse communities. However, this connectivity has also accelerated the spread of harmful language. Online environments, often characterised by anonymity, embolden users to engage in aggressive or hateful behaviour that is typically avoided in physical interactions. Harmful language ranges from general profanity to targeted hate speech designed to incite violence or discriminate against marginalised groups.

The volume of daily content makes manual moderation unscalable. Traditional automated methods often rely on simple lexicons that fail to interpret context, irony, or the reclaimed use of slurs. This leads to the "offensive language problem," where neutral or merely crude posts are misclassified as hate speech. The core challenge is developing context-aware models that accurately distinguish between hate speech, offensive language, and neutral content. This research explores a hybridised approach combining unsupervised and supervised learning to address these classification nuances.

2. Research Objective

The primary objective of the research is to design and test a hybrid machine learning system capable of identifying harmful language on social media with high accuracy. The combination of unsupervised exploration and supervised prediction that we hope to achieve, being unable to confuse classes, will appear typical with single-model solutions.

The goal is to create a multi-classification system that indicates posts as hate speech, offensive language, or simply safe material. Further, the study explores linguistic peculiarities and theme groups to understand how these toxic messages manifest online. Eventually, we want to contribute to the creation of larger, more equitable content moderation systems with smart, cluster-based functionality.

3. Dataset Description

In this research, the dataset used is the list of hate speech and offensive language as provided by Davidson et al. (2017). The dataset includes about 24,783 tweets gathered using a hate-speech lexicon. At least three human coders manually annotated each tweet and assigned it a category based on the message's context. The data is typified by high class imbalance, as reflected in reality.

Approximately 5% of tweets are labelled as hate speech, 77% as offensive language, and 17% as neither. The main variables are the number of CF users across different

categories, the final class label (0 = hate, 1 = offensive, 2 = neutral), and the uncoded tweet texts.

4. Methodology

4.1 Data Processing

To prepare the X (formerly 'Twitter') data for my project, I am establishing a multi-stage preprocessing pipeline. The raw X-data is exceedingly noisy, sprinkled with handles for accounts, web addresses, and various platform-specific symbols.

The process includes:

4.1.1 Lowercase and Stripping Non-English: Rewrite everything in lowercase and remove characters that are not English.

4.1.2 Noise Removal: It removes URLs, hashtags, and punctuation with the help of a regular expression (regex).

4.1.3 Tokenisation and Stop Word Filtering: Divide the tweets into single words and remove common and meaningless words such as the or is.

4.1.4 Stemming: Use a Porter Stemmer whereby words are cut down to the language base (e.g. killing is cut down to kill).

4.2 TF-IDF Vectorisation:

Term Frequency-Inverse Document Frequency (TF-IDF) is used to convert a processed text into numerical vectors. Basically, the model assigns higher weights to words that appear in a specific tweet but are quite uncommon across the entire dataset. This helps the system focus on the most meaningful offensive words while ignoring typical linguistic filler.

4.3 Unsupervised Learning: K-Means Clustering

The K-Means clustering is initially applied to the tweets that have already been vectorised to detect latent thematic structures without the aid of class labels. The algorithm iteratively groups data points into clusters k minimising the Euclidean distance between the data points and their cluster centres. In this context, K-Means finds clusters of similar semantic signatures of tweets - e.g., racial slurs cluster, gender-based profanity cluster, or neutral everyday. The cluster label of each resulting tweet is added as an extra feature to the input matrix for the supervised classifiers.

4.4 Supervised Learning: Classification Models

The augmented dataset is used to train supervised models. This study focuses on two primary classifiers:

4.4.1 Naive Bayes (NB): A probabilistic method of classification that is based on the Bayes Theorem, and which assumes that features do not actually depend on each other. It is quick to execute and performs well with large volumes of text data.

4.4.2 Logistic Regression (LR): It is a linear model that computes the odds of each class using a sigmoid function. It is typically the default option in multi-class frameworks because it can approximate which features are most important and is relatively effective with noisy inputs.

5. Data Visualisation and Analysis

5.1 Class Distribution and Word Clouds

Word clouds are used to distinguish different linguistic profiles.

Hate Speech: This is usually written by slurs that are identity-based and are directed towards race, religion and sexual orientation.

Offensive Language: This is typified by profanity, such as 'bitch' or 'hoe', that implies being outside the in-group.

Neither: Neutral descriptive terms are present, but some identity terms (e.g., black, white) can be used as well, but in a non-offensive manner.

5.2 Cluster Analysis

The Principal Component Analysis (PCA) projection of the vectors into a two-dimensional space shows that the Offensive Language category often serves as a semantic intermediate point between the categories neutral and hateful. Cluster analysis is used to design subcategories within the offensive category, such as categorising aggressive banter and dehumanising language.

5.3 Model Evaluation

Accuracy, precision, recall, and the F1-score, calculated from a confusion matrix, are used to evaluate the model's performance. The evaluation should be conducted per-class because the dataset is imbalanced; a high overall accuracy can be misleading if the model does not detect the minority class, the Hate Speech one. The confusion matrix typically shows that when hate speech is misidentified as offensive language, the most common mistake is conflating the two terms because they share similar words. Hybrid models can be more accurate than single classifiers, and the accuracies of Logistic Regression augmented with K-Means features are often more than 85-90 per cent.

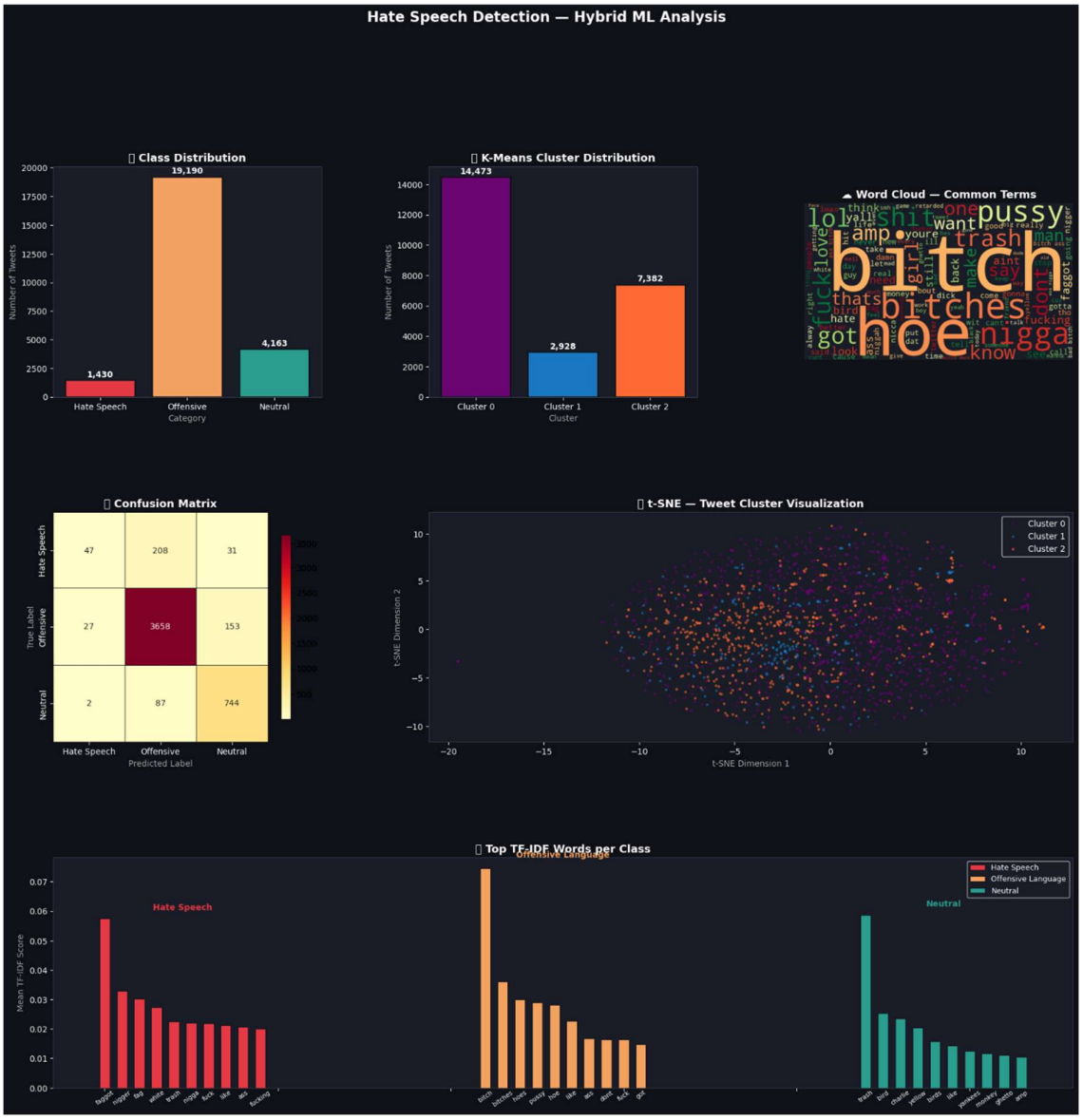
6. Social Impact and Implications

The moderation systems are based on hybrid machine-learning techniques that filter for explicit toxicity at scale, protecting users from psychological damage. By automatically defusing cases of unequivocal violations, such systems allow human moderators to focus on borderline cases. Effective detection also reduces the chilling effect that online harassment has on marginalised groups**.**

7. Conclusion

This paper demonstrates the effectiveness of a machine-learning hybrid method for detecting hate speech. The method provides important contextual information that improves classification accuracy by combining unsupervised K-Means clustering with supervised Logistic Regression or Naive Bayes classifiers. Despite improvements to the hybrid model in detection, reducing algorithmic bias remains a concern on the path to inclusive digital environments. Finally, such systems represent a major breakthrough in ensuring safe and respectful online communication.

DATA VISUALS



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